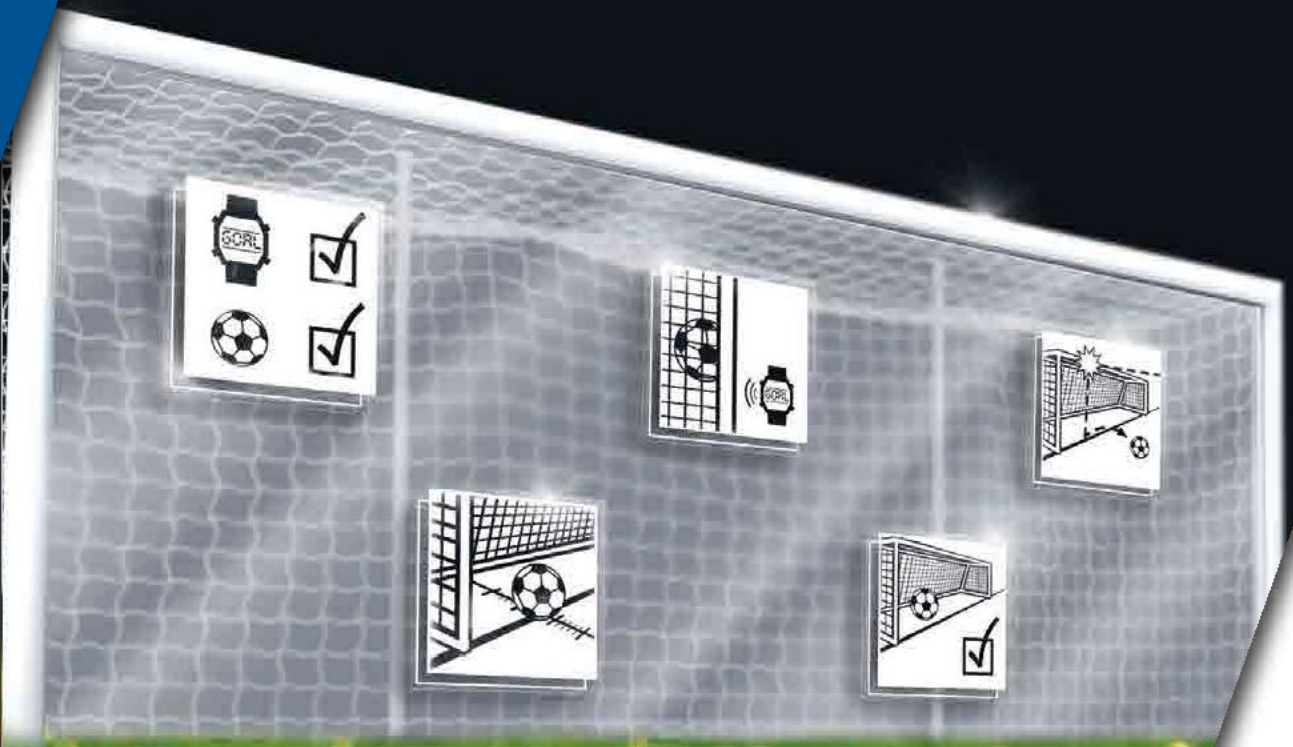
FIFA QUALITY PROGRAMME

FOR GOAL-LINE TECHNOLOGY

TESTING MANUAL 2014,
2019 Update





FIFA® QUALITY

FIFA Quality Programme for Goal-Line Technology

TESTING MANUAL 2014

2019 Update

CONTENTS

	Page
A Introduction	**5**
1 Goal-line technology	6
2 Scope of the manual	7
3 Purpose of the manual	7
4 Terms and definitions	7
B Licensing and certification	**9**
1 Licensing procedure	10
2 Certification process	10
3 Types of tests	11
4 Test requirements	11
5 Test methods	23
6 Maintenance	41
7 Warranty	41
C Check by the referee	**43**
1 Tests before the match	44
2 Provision of information to competition organiser	46
Appendix A CAD drawing of target wall	47
Appendix B CAD drawing of slider	48

While every effort has been made to ensure the accuracy of the information contained in this testing manual, any party who makes use of any part of this manual in the development of a goal-line technology system (a "User") does so at its own risk and FIFA shall have no liability with regard to such use. Any use of this manual including by undergoing any test of a system by a User constitutes acceptance of the terms of this disclaimer by that User. The GLT test procedure and this testing manual are FIFA's intellectual property. FIFA reserves the right to amend, update or delete sections of this manual at any time as it deems fit.


adidas
FIFA WORLD CUP
Brasil



INTRODUCTION

1 GOAL-LINE TECHNOLOGY

1.1 Objective

The objective of goal-line technology (GLT) is not to replace the role of the officials, but rather to support them in their decision-making. The technology must provide a clear indication as to whether the ball has fully crossed the line, and this information will serve to assist the referee in taking his final decision.

1.2 Review

At its Annual Business Meeting on 20 October 2010, the International Football Association Board (IFAB) discussed the implications of measurement systems that are capable of automatically detecting the scoring of a goal during a football match (so-called goal-line technology or GLT). IFAB laid down a set of four basic requirements a GLT system has to fulfil:

i) The goal-line technology applies solely to the goal line and only to determine whether a goal has been scored or not;
ii) The GLT system must be accurate;
iii) The indication of whether a goal has been scored must be immediate and automatically confirmed within one second; and
iv) The indication of whether a goal has been scored will be communicated only to the match officials (via the referee's watch, by vibration and visual signal).

After preliminary tests at the Home of FIFA in Zurich in February 2011, FIFA and EMPA (an independent Swiss test institute), in cooperation with IFAB, performed test phase 1 in November and December 2011.

Phase 1: general evaluation of the system

Phase 2: evaluation of the reliability and robustness of the GLT system, including in-depth analysis of the GLT system's accuracy with the aim of testing the various GLT systems to their limits

Based on these independent test results from phases 1 and 2, IFAB decided on 5 July 2012 that the potential use of goal-line technology was to be implemented in the Laws of the Game and a certification procedure should control the quality of GLT systems.

This document is based on the experience of the testing phases in 2011/2012 and the first tournaments with GLT in 2012 and 2013. Small updates have now been added after consultation with the test institutes involved in the certification process since the programme's inception. The defined requirements and test methods should therefore ensure high-quality goal-line technology in the world of football.

2 SCOPE OF THE MANUAL

This testing manual has been developed by FIFA with the objective of defining test procedures for GLT systems that are based on the requirements set by IFAB and applicable and valid worldwide. Based on these principles, this manual describes the test procedure for a GLT system to test its functionality, determine its accuracy and evaluate its robustness. The GLT test procedure is FIFA's intellectual property and refers to football only.

3 PURPOSE OF THE MANUAL

This testing manual contains all the information that is needed to test a goal-line technology system for the GLT system test, final installation test in a stadium and the referee checks required for final approval before the match.. It may only be used by FIFA-accredited test institutes for goal-line technology

4 TERMS AND DEFINITIONS

This testing manual incorporates by dated or undated reference provisions from other publications. For dated references, subsequent amendments to or revisions of any of these publications will apply to this manual only when incorporated into it by amendment or revision. For undated references, the latest edition of the publication referred to applies.

- Laws of the Game (www.fifa.com/aboutfifa/officialdocuments/doclists/laws.html)
- FIFA Quality Programme for Footballs (https://football-technology.fifa.com/en/standards/footballs/)
- BS EN 748:2013+A1:2018 - Playing field equipment. Football goals. Functional and safety requirements, test methods (www.cen.eu)
- IEC 60825 Laser Safety Instructions and Regulations (www.cen.eu)
- IEC 60068 Environmental Testing (www.cen.eu)
- IEC 61000-4-2 Electromagnetic compatibility (ww.cen.eu)


B
01
1



LICENSING AND CERTIFICATION

1 LICENSING PROCEDURE

1.1 Application

Providers of GLT systems are invited to apply for the FIFA licence. In order to guarantee a high level of quality for the end user, GLT providers must prove that they meet certain requirements in terms of system production, business management and social responsibility. These requirements are defined in the application document on https://football-technology.fifa.com/en/

1.2 GLT system test

In order to become a FIFA licensee, the GLT system test, which is conducted by a FIFA-accredited test institute, must be passed (a list of which can be found on https://football-technology.fifa.com/) . This involves subjecting the system to an intensive test programme in the laboratory, on the field and in simulated real game situations.

1.3 FIFA licence

If the application and test criteria are met, FIFA will issue a standard licence agreement to the GLT provider. On signing the contract, the GLT provider is then licensed to make agreements with customers and to install its GLT system. The name of the GLT provider as a licensee and the system will be publicly accessible on https://football-technology.fifa.com/en/.

2 CERTIFICATION PROCESS

2.1 Installation of the system

The FIFA licence authorises the GLT provider to install the GLT system anywhere in the world. The GLT provider must ensure that the end user is properly informed about the technology and how to operate and maintain the system and must provide each end user with a maintenance guide. It must also provide the end user with a warranty of at least two years for its GLT system (see B7).

2.2 Final installation test

On completion of an installation and in order to be allowed to use the installed GLT system in official matches, each individual installed GLT system must be tested again by a FIFA-accredited test institute. The final GLT installation test is a modified version of the GLT system test, the aim being to test the functionality of each GLT system following installation.

2.3 Acceptance of the installation

The installation and results of the final installation test must be accepted by the GLT provider's customer (e.g. competition organiser, stadium owner, club, etc.) using FIFA's standard acceptance and acknowledgement form. The signed document must then be submitted to FIFA for final approval and sign-off.

2.4 FIFA QUALITY PRO certification

On final approval of the documents by FIFA, the installation is awarded the FIFA QUALITY PRO seal and listed on https://football-technology.fifa.com/en/. The GLT system can now be used in official matches/competitions for a period of 12 months and must be re-tested annually.

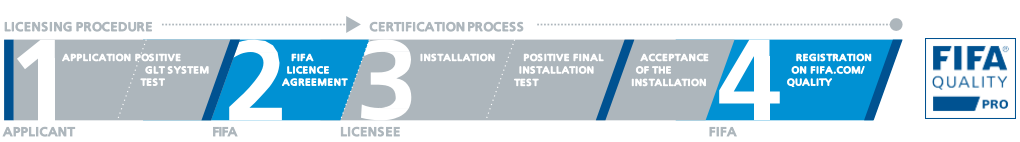


Figure 1: Four steps to a FIFA QUALITY PRO certification

3 TYPES OF TESTS

To ensure the highest possible quality of goal-line technology, any approved GLT system must successfully pass a series of system tests as well as a final installation test before it is approved for use in official matches (such matches to which the Laws of the Game apply). Therefore, two different test programmes have been developed which are applied in the licensing and certification process.

3.1 GLT system test

Consisting of laboratory tests, field tests and real game situations in training sessions, the purpose of these system tests is to give competition organisers, stadium owners, clubs, etc. intending to install GLT systems the assurance that the system (subject to the final installation test, see section 3.2 below) complies with the Laws of the Game.

These tests have to be conducted by an independent FIFA-accredited test institute. While the purpose of the laboratory tests is to examine the components of each technology, the field tests and tests under real game conditions aim primarily at evaluating their functionality, accuracy and robustness. Only if the system fulfils all the necessary criteria will it be licensed and registered by FIFA as a licensed GLT system (https://football-technology.fifa.com/en/).

The GLT system test is divided into three stages:

- Laboratory tests: testing of balls, football goals, referee's watches and other components of the system
- Field tests in two stadiums (the tests will be conducted on both goals): general evaluation of the system and analysis of its accuracy
- Training session: tests under real game conditions

3.2 Final installation test

This final step in the process verifies the proper installation of the licensed GLT system in a given stadium. The final installation test aims to establish the functionality of the system by certifying that the technology performs to the same level once installed in a given stadium as it did during the system test. A modified protocol of tests according to the GLT system test is applied. Following a successful final installation test, approval by the customer (e.g. competition organisers, stadium owners, clubs, etc.) and the final approval of the documents by FIFA, the GLT system will then appear on the official FIFA list of certified installations (https://football-technology.fifa.com/en/).

4 TEST REQUIREMENTS

This section describes the requirements that are used to assess the accuracy, functionality and robustness of a GLT system.

The test method of each test is defined and described in detail in section 5 of this manual.

For any field test (system test or final installation test), the setup of the goal-line technology system must be in accordance with future match conditions (with or without floodlights) in the respective venue. This includes all items such as the original goal frames, the correct nets and net tension as well as lighting conditions and balls in accordance with the FIFA Quality Programme for Footballs. The ball shall be inflated to 0.8 ±0.05 bar.

The testing schedules are divided into different tests. The tests should be randomised at the discretion of the test institute. As a general rule, no adjustments, calibrations, adaptations are to be made by the provider during the tests. The provider

may make a request for changes with the test institute. The test institute may allow such changes but will inform the provider of any tests that may need repeating with new settings. Other than this no communication should be had between the provider and test institute during testing.

Any "GOAL" indication or other unusual activity on the referee watch that occurs outside of the compulsory testing blocks is mandatorily to be documented in the report by the test institute.

4.1 GLT system test

In order to become a FIFA licensee, the GLT system test, which is conducted by a FIFA-accredited test institute, must be passed. This involves subjecting the system to an intensive test programme in the laboratory, on the field and in simulated real game situations. The table below gives an overview of the requirements that the GLT system has to fulfil in order to obtain a FIFA licence.

	TEST	TEST METHOD	SPECIFICATION	TEST CONDITIONS	SHOTS	REQUIREMENTS	GLT SYSTEM TEST
1	Test of footballs	FIFA GLT TEST 01	-	Laboratory	-	FIFA Quality/Quality Pro or IMS standard	√
2	Test of goals	FIFA GLT TEST 02	Safety	Laboratory (field)	-	BS EN 748:2013+A1:2018 and additional evidence by an independent test institute	√
			Functionality			BS EN 748:2013+A1:2018	
			Influence			Evidence by an independent test institute	
3	Test of durability of referee's watch	FIFA GLT TEST 03	Climatic	Laboratory	-	EN 60068	√
			Mechanical			Functionality after tests	
			Electromagnetic			EN 61000-4-2	
			Durability		50	>5h	
4	Laboratory test of critical components	FIFA GLT TEST 04	Defined by the test institute	Laboratory	-	-	√
5	Unrestricted shots	FIFA GLT TEST 05	-	Field	55	Correct indication of all shots during two sets of tests	√
6	Tests with target wall	FIFA GLT TEST 06	Free view	Field	42	Accuracy ± 1.5cm during two sets of tests	√
			Restricted view	Field	10	Accuracy ± 1.5cm for all shots	√
7	Slider test	FIFA GLT TEST 07	-	Field	20	Accuracy ± 1.5cm during two sets of tests	√
8	Tests of functionality of referee's watch	FIFA GLT TEST 08	Coverage	Field	10	Full pitch with run-offs during two sets of tests	√
9	Additional ran-domised shots on empty goals	FIFA GLT TEST 09	-	Field	25	Correct indication of all shots during two sets of tests	√
10	Training session	FIFA GLT TEST 12	-	Field	40	Correct indication of all shots	√
11	Tests of immunity to adverse conditions	FIFA GLT TEST 11	-	Field	50	ADDITIONAL INFORMATION ONLY	√
12	Goalkeeper test	FIFA GLT TEST 10		Field	30	Correct indication of all shots	√
13	Delay of referee's watch	FIFA GLT TEST 13	Goal indication		27	≤1second during two sets of tests	√

Table 1: Requirements of the GLT system test

4.1.1 Laboratory tests

Most of the test requirements for the laboratory tests are based on international (ISO, EN etc.) or FIFA standards (see A4). All tests have to be passed.

4.1.1.1 Test of footballs

The international requirements for footballs that can be used in official matches are defined in the handbook of test requirements for footballs (see A4). All types of balls that are produced or modified by the GLT provider (technology inside the ball) have to be tested according to the "FIFA QUALITY", "FIFA QUALITY PRO" or "IMS International Matchball Standard" standard (or any marks that supersede these) by a FIFA-accredited test institute for footballs. Seven tests are carried out in order to get the "FIFA QUALITY PRO" seal and six test have to be passed for the "FIFA QUALITY" or the "IMS International Matchball Standard".

Any ball that includes components of the technology must be fully functional both before and after the tests of the FIFA Quality Programme for Footballs. This is especially in reference to the shape and size retention test during which the ball is shot against a rigid surface 2000 times.

Once a ball is tested at a stadium it may be used at any stadium with the same installed system for a period of 1 year. Each time a FIT happens the expiry date of that ball is extended by 1 year. Providers may keep track of this in any way they see fit, but are responsible for this process and the necessary communication with the labs. The test institute is to ensure that for every GLT test report the correct ball model is added using the official name of the ball – this way we have a backup of the information

4.1.1.2 Test of goals (i.e. the goal itself)

The requirements are based on the BS EN 748:2013+A1:2018 standard for playing equipment (see A4). In addition, the FIFA-accredited test institute has to decide after the analysis of the technology and the modification of the goal if further tests are needed. The proposal for additional tests has to be submitted to FIFA, whose written approval is needed. The requirements of BS EN 748:2013+A1:2018 standard and all additional tests have to be passed.

4.1.1.3 Test of durability of referee's watch

The testing of the durability of the referee's watch consists of four laboratory tests. All three watches, which have to be provided to the FIFA-accredited test institute by the provider of the GLT, have to meet the defined requirements. These are based on the EN 60068 (see A4), EN 61000-4-2 (see A4) standards and a developed test for the functionality (see section 5.1.3.3). The battery of the watches must last more than five hours with a "GOAL" indication every 5 minutes. Each goal indication consists of a visual and vibration signal for between 10 and 20 seconds.

4.1.1.4 Laboratory test of critical components

The components of a GLT system are to be identified by the FIFA-accredited test institute. Based on this analysis and the nature of the relevant GLT system and the method it uses for determining if the ball has crossed the goal line, the test institute and FIFA decide on the appropriate tests and requirements for those components. Preference is to be given to internationally recognised tests. The aim of this section is to prevent technology that has obvious flaws that are not taken into account by the current test methods. A costed testing proposal should be recommended to FIFA by the test institute and subject to FIFA's approval, the testing can take place at a cost to the provider.

4.1.2 Field tests

The field test will be performed by the FIFA-accredited test institute at two different stadiums on both goals. The provider of the GLT system will choose the two stadiums where the field test is to be carried out (each such stadium is subject to FIFA's prior written approval). As part of the field test, the GLT system will undergo a set of tests where the GLT system will be evaluated on the basis of five individual requirements (listed in the following

sections). In order to pass the field test, the GLT system must pass each of the above-mentioned five individual requirements at each of the two stadiums.

Each Block has to pass fully without failure, upon failure the block can be repeated with double the number of events as the original block until it passes, the number of block failures must be recorded in the test report. A block is defined by the letter of the event, with the exception of the slider test (E), here E1/E2 is one block and E3/E4 is a second block. Any testing that requires departing a stadium and returning on a different day, requires testing to be started from the beginning.

The individual requirements that the GLT system has to pass during the field test according to the above are the following:

4.1.2.1 Unrestricted shots

All shots in the goal and near misses have to be correctly indicated on the watch. A detailed description of each shot A01-A07 and D01-D02 can be found in section 5 of this testing manual. Two different tests are carried out:
a) Unrestricted shots (goal or near misses)
b) Unrestricted shots into the outer netting (no goal)

UNRESTRICTED SHOTS	DESCRIPTION OF TEST	SHOTS	REQUIREMENT
A01	Centre, flat 35km/h	2	Goal
A02	Centre, high 70km/h	2	Goal
A03	Left, flat 70km/h	2	Goal
A04	Left, high 35km/h	2	Goal
A05	Right, flat 35km/h	2	Goal
A06	Right, high 70km/h	2	Goal
A07	Shots against the post 70km/h	2	Goal or No Goal

Table 2: a) Unrestricted shots (GLT system test)

UNRESTRICTED SHOTS INTO THE OUTER NETTING	DESCRIPTION OF TEST	SHOTS	REQUIREMENT
D01	Over the crossbar (manually)	2	No goal
D02	Into either side netting (manual-ly, varying speeds)	2	No goal

Table 2: b) Unrestricted shots into the outer netting (GLT system test)

4.1.2.2 Tests with target wall

All shots have to be correctly indicated on the watch with an accuracy of ± 1.5cm. A detailed description of each shot B01-B07 and C01-C02 can be found in section 5 of this testing manual. This test consists of two different test procedures:

a) Target wall with unrestricted view
b) Target wall with restricted view

TARGET WALL	POSITION OF WALL	SHOTS	REQUIREMENT
B01	120mm behind goal line, all B shots at 50km/h	12	No goal
B02	160mm behind goal line	12	To be defined by HS camera
B03	200mm behind goal line	12	To be defined by HS camera
B04	240mm behind goal line	12	Goal
B05	to be defined by test institute based on B01-B04	12	To be defined by HS camera
B06	to be defined by test institute based on B01-B04	12	To be defined by HS camera
B07	to be defined by test institute based on B01-B04	12	To be defined by HS camera

Table 3: a) Target wall with unrestricted view (GLT system test)

TARGET WALL WITH RESTRICTED VIEW	POSITION OF WALL	SHOTS	REQUIREMENT
C01	120mm behind goal line 35km/h	5	No goal
C02	240mm behind goal line 70 km/h	5	Goal

Table 3: b) Target wall with restricted view (GLT system test)

4.1.2.3 Slider test

Each test has to be correctly indicated on the watch with an accuracy of ± 1.5cm from the 0. A detailed description of the two test scenarios E01-E02 can be found in section 5 of this testing manual.

SLIDER TEST	POSITION OF SLIDER	SHOTS	REQUIREMENT
E01	Left side of the goal	5	± 1.5cm
E02	Centre of the goal	5	± 1.5cm

Table 4: Slider test (GLT system test)

4.1.2.4 Tests of functionality of referee's watch

The referee's watch has to cover the entire pitch including the area which is at least 5 metres outside the perimeter of the pitch (or as far as possible if the runoff is smaller than 5 metres). In addition, the "GOAL" signal has to be indicated on the watch within one second. The results are to be rounded to integers and must be ≤ 1 second. The two different test scenarios U01-U02 and the procedure used for the calculation of the delay of the referee's watch are described in section 5 of this testing manual.

a) Coverage of referee's watch
b) Indication on the referee's watch within one second

COVERAGE OF THE REFEREE'S WATCH	DESCRIPTION OF TEST	SHOTS	REQUIREMENT
U01	Corner, centre, side line	5	Goal
U02	as for U01 but with senders	5	Goal

Table 5: a) Coverage of the referee's watch (GLT system test)

FUNCTIONALITY	DESCRIPTION OF TEST	SHOTS	REQUIREMENT
A01	Unrestricted shots	5	≤ 1.0 s
B01	Target wall	12	≤ 1.0 s
F01	Goalkeeper test	5	≤ 1.0 s

Table 5: b) Indication on the referee's watch within one second (GLT system test)

4.12.5 Additional randomised shots

All shots have to be correctly indicated on the watch. The test procedure for the shots B01-E02 is described in detail in section 5 of this testing manual.

a) With ball

RANDOMISED SHOTS	DESCRIPTION OF TESTS	SHOTS	REQUIREMENT
B01	Target wall	1	Goal
B02	Target wall	1	Goal
B03	Target wall	1	Goal
B04	Target wall	1	Goal
B05	Target wall	1	Goal
B06	Target wall	1	Goal
B07	Target wall	1	Goal
C01	Target wall	1	Goal
C02	Target wall	1	Goal
D01	Unrestricted shots	1	Goal
D02	Unrestricted shots	1	Goal
E01	Slider test	1	Goal
E02	Slider test	1	Goal
E03	Slider test	1	Goal
E04	Slider test	1	Goal

Table 6 a): Additional randomised shots (GLT system test)

b) With non-ball projectiles (as described in 5.2.5.2)

RANDOMISED SHOTS	DESCRIPTION OF TEST	SHOTS	REQUIREMENT
B01	Target wall	1	No Goal
B02	Target wall	1	No Goal
B03	Target wall	1	No Goal
B04	Target wall	1	No Goal
B05	Target wall	1	No Goal
B06	Target wall	1	No Goal
B07	Target wall	1	No Goal
C01	Target wall	1	No Goal
C02	Target wall	1	No Goal
D01	Unrestricted shots	1	No Goal
D02	Unrestricted shots	1	No Goal
E01	Slider test	1	No Goal
E02	Slider test	1	No Goal
E03	Slider test	1	No Goal
E04	Slider test	1	No Goal

Table 6 b): Additional randomised shots with non-ball projectiles (GLT system test)

4.1.2.6 Tests of immunity to adverse conditions

These tests are not pass/fail criteria to obtain the FIFA licence but give additional information for the provider of the GLT system. The detailed test procedure for G01-G09 is defined in section 5 of this testing manual.

ADVERSE CONDITION	DESCRIPTION OF TEST	SHOTS	REQUIREMENT
G01	Bright objects	5	Goal
G02	Snow simulation	5	Goal
G03	Dirty ball	5	Goal
G04	Laser pointer / field distortion	5	Goal
G05	Flashlight	5	Goal
G06	Smoke test	5	Goal
G07	Interfering transmitter	5	Goal
G08	2nd ball	5	Goal
G09	Line test	5	Goal
G10	Non-ball projectiles	5	No Goal

Table 7: Tests of immunity to adverse conditions (GLT system test)

4.1.2.7 Goalkeeper test

All shots have to be correctly indicated on the watch. The description of these tests can be found in section 5.2.6 of this testing manual.

GOALKEEPER TEST	DESCRIPTION OF TEST	SHOTS	REQUIREMENT
F01	Catching in hands	5	Goal
F01	Catching in hands (without ball)	5	No goal
F02	Catching in arms	5	Goal
F02	Catching in arms (without ball)	5	No goal
F03	Rolling through tunnel	5	Goal
F03	Rolling through tunnel (without ball)	5	No goal

Table 8: Goalkeeper test (GLT System test)

4.1.3 Real game situations in training session

A training session is conducted in order to simulate real game situations. Tests H01-H08, which are defined in section 5 of this testing manual, are carried out. All situations have to be correctly indicated on the watch. Due to the importance of the functionality in real game situations, this test is conducted only once.

TRAINING SESSION	DESCRIPTION OF TEST	SHOTS	REQUIREMENT
H01	Goal 1, corner right	5	Goal
H02	Goal 1, corner left	5	Goal
H03	Goal 1, free kick, Pos. 1	5	Goal
H04	Goal 1, free kick, Pos. 2	5	Goal
H05	Goal 2, corner right	5	Goal
H06	Goal 2, corner left	5	Goal
H07	Goal 2, free kick, Pos. 1	5	Goal
H08	Goal 2, free kick, Pos. 2	5	Goal

Table 9: Training session (GLT system test)

4.1.4 System test period

The tests listed in section 4.1.1 – 4.1.3 have to be carried out over a maximum of a 12-month period. Upon expiry of this 12-month period, the GLT system has to pass the entire system test again.

4.2 Final installation test

Once the GLT provider has been appointed as a FIFA licensee, having passed the tests described in 4.1.1 – 4.1.3, each given installation of a GLT system has to pass a final installation test again. In this regard, on completion of an installation, the GLT system must be tested again by a FIFA-accredited test institute The final installation test is a modified version of the system test, the aim being to test the functionality of the system on both goals in the relevant stadium.
The table below gives an overview of the requirements that the installation has to fulfil in order to obtain FIFA certification. The final installation test will consist of one set of tests consisting of eight criteria. Each of the criteria must be passed. Please refer to the procedural rules stated in section 4 for test conditions.

	TEST	TEST METHOD	SPECIFICATION	SHOTS	REQUIREMENTS	INSTALLATION TEST	GOAL
1	Unrestricted shots	FIFA GLT TEST 05	-	55	Correct indication of all shots	√	A
2	Tests with target wall	FIFA GLT TEST 06	Free view	36	Accuracy ± 1.5cm of all shots	√	B
			Restricted view	10	Accuracy ± 1.5cm of all shots	√	B
			Restricted view indirect	10	Accuracy ± 1.5cm of all shots	√	B
3	Slider test	FIFA GLT TEST 07	-	20	Accuracy ± 1.5cm of all shots	√	A&B
4	Tests of functionality of referee's watch	FIFA GLT TEST 08	Coverage	10	Full pitch with run-offs of all shots	√	A
5	Additional randomised shots on empty goals	FIFA GLT TEST 09	Ball	13	Correct indication of all shots	√	A&B
			Non-ball projectiles	13			
6	Goalkeeper test	FIFA GLT TEST 10	-	30	Correct indication of all shots	√	A
7	Product identification	-	-	-	Same components of the GLT system as identified in FIFA GLT TEST 04	√	-
8	Delay of referee's watch	FIFA GLT TEST 13	Goal indication	27	≤1second during of all shots	√	A&B

Table 10: Requirements of the final installation test

4.2.1 Field test

Both goals are required for the field tests. Some tests are carried out on both goals, others only on one in accordance with this testing protocol. The installed GLT system must pass all these tests.

Each venue shall be tested in compliance with this manual with the primary competition ball that will be used. In order for further balls to be eligible to be used in a given venue, every further ball shall in addition fulfil the following tests in accordance with the requirements of the final installation test:

- Test with target wall (FIFA GLT TEST 06)
- Slider test (FIFA GLT TEST 07)
- Goalkeeper test (FIFA GLT TEST 10)

4.2.1.1 Unrestricted shots

All shots in the goal and near misses have to be correctly indicated on the watch. A detailed description of each shot A01-A07 and D01-D02 can be found in section 5.2.1 of this testing manual. Two different tests are carried out:

a) Unrestricted shots (goal or near misses)
b) Unrestricted shots into the outer netting (no goal)

UNRESTRICTED SHOTS	DESCRIPTION OF TEST	SHOTS	REQUIREMENT
A01	Centre, flat 35km/h	2	Goal
A02	Centre, high 70km/h	2	Goal
A03	Left, flat 70km/h	2	Goal
A04	Left, high 35km/h	2	Goal
A05	Right, flat 35km/h	2	Goal
A06	Right, high 70km/h	2	Goal
A07	Shots against the post 70 km/h	2	Goal or No goal

Table 11: a) Unrestricted shots (Final installation test)

UNRESTRICTED SHOTS INTO THE OUTER NETTING	DESCRIPTION OF TEST	SHOTS	REQUIREMENT
D01	Over the crossbar (manually), landing on top of the net	2	No goal
D02	Into either side netting (manually, varying speeds)	2	No goal

Table 11: b) Unrestricted shots into the outer netting (Final installation test)

4.2.1.2 Tests with target wall

All shots have to be correctly indicated on the watch with an accuracy of ± 1.5cm. A detailed description of each shot B01-B03 and C01-C04 can be found in section 5.2.2 of this testing manual. This test consists of three different test procedures:

a) Target wall with unrestricted view
b) Target wall with restricted view

TARGET WALL	POSITION OF WALL	SHOTS	REQUIREMENT
B01	180mm behind goal line, all B shots 50km/h	12	To be defined by HS camera
B02	15mm behind exact goal position of target wall	12	Goal
B03	15mm in front of exact goal position of target wall	12	No Goal

Table 12: a) Target wall with unrestricted view (Final Installation test)

TARGET WALL WITH RESTRICTED VIEW	POSITION OF WALL	SHOTS	REQUIREMENT
C01 (35 km/h)	120mm behind goal line	5	No goal
C03 (shots by player)	120mm behind goal line	5	No goal
C02 (70 km/h)	240mm behind goal line	5	Goal
C04 (shots by player)	240mm behind goal line	5	Goal

Table 12: b) Target wall with restricted view (Final installation test)

4.2.1.3 Slider test

Each test has to be indicated correctly on the watch with an accuracy of ± 1.5cm from the 0. A detailed description of the two test scenarios E01-E02 can be found in section 5.2.3 of this testing manual. The test is to be performed on both goals.

SLIDER TEST	POSITION OF SLIDER	SHOTS	REQUIREMENT
E01	Side of the goal	5	± 1.5cm
E02	Centre of the goal	5	± 1.5cm
E03	Side of the goal	5	± 1.5cm
E04	Centre of the goal	5	± 1.5cm

Table 13: Slider test (Final installation test)

4.2.1.4 Testing of functionality of referee's watch

The referee's watch has to cover the entire pitch including the area which is at least five (5) metres outside the perimeter of the pitch. In addition, the "GOAL" signal has to be indicated on the watch within one second. The results are to be rounded to integers. The two different test scenarios U01-U02 and the procedure used for the calculation of the delay of the referee's watch are described in section 5 of this testing manual.

COVERAGE OF THE REFEREE'S WATCH	DESCRIPTION OF TEST	SHOTS	REQUIREMENT
U01	Corner, centre, sideline	5	Goal
U02	as for U01 but with senders	5	Goal

Table 14: Coverage of the referee's watch (Final installation test)

In addition the delay of goal indication is to be measured on 27 selected shots. For these shots, the footage from the HD camera placed on the goal-line is to be evaluated. All 27 shots must be correctly indicated as "GOAL" within 1 second.

FUNCTIONALITY	DESCRIPTION OF TEST	SHOTS	REQUIREMENT
A01	Unrestricted shots	5	≤ 1.0 s
B02	Target wall	12	≤ 1.0 s
F01	Goalkeeper test	5	≤ 1.0 s

Table 15: Indication on the referee's watch within one second (Final installation test)

4.2.1.5 Additional randomised shots on empty goals

All shots have to be correctly indicated on the watch. The test procedure for the shots B01-E02 is described in detail in section 5.2.5 of this testing manual.

a) With ball

RANDOMISED SHOTS	DESCRIPTION OF TEST	SHOTS	REQUIREMENT
B01	Target wall	1	Goal
B02	Target wall	1	Goal
B03	Target wall	1	Goal
C01	Target wall	1	Goal
C02	Target wall	1	Goal
C03	Target wall	1	Goal
C04	Target wall	1	Goal
D01	Unrestricted shots	1	Goal
D02	Unrestricted shots	1	Goal
E01	Slider test	1	Goal
E02	Slider test	1	Goal
E03	Slider test	1	Goal
E04	Slider test	1	Goal

Table 16 a): Additional randomised shots (Final installation test)

b) With non-ball projectiles (defined in 5.2.5.2)

RANDOMISED SHOTS	DESCRIPTION OF TEST	SHOTS	REQUIREMENT
B01	Target wall	1	No Goal
B02	Target wall	1	No Goal
B03	Target wall	1	No Goal
C01	Target wall	1	No Goal
C02	Target wall	1	No Goal
C03	Target wall	1	No Goal
C04	Target wall	1	No Goal
D01	Unrestricted shots	1	No Goal
D02	Unrestricted shots	1	No Goal
E01	Slider test	1	No Goal
E02	Slider test	1	No Goal
E03	Slider test	1	No Goal
E04	Slider test	1	No Goal

Table 16 b): Additional randomised shots with non-ball projectiles (Final installation test)

4.2.1.6 Goalkeeper test

All shots have to be correctly indicated on the watch. The description of these tests can be found in section 5.2.6 of this testing manual.

GOALKEEPER TEST	DESCRIPTION OF TEST	SHOTS	REQUIREMENT
F01	Catching in hands	5	Goal
F01	Catching in hands (without ball)	5	No goal
F02	Catching in arms	5	Goal
F02	Catching in arms (without ball)	5	No goal
F03	Rolling through tunnel	5	Goal
F03	Rolling through tunnel (without ball)	5	No goal

Table 17: Goalkeeper test (Final installation test)

4.2.2 Product identification

The components of the installed GLT system have to comply with the product identification in FIFA GLT TEST 05 (see 5.1.4).

For camera-based systems, the serial numbers of all cameras and the software version shall be obtained from the provider and noted in the report.

For systems using electro-magnetic fields, the software version, serial number of the goals and of the computer rack shall be obtained from the provider and noted in the report.

	TEST	TEST METHOD	SPECIFICATION	TEST CONDITIONS	SHOTS	REQUIREMENTS	GLT SYSTEM TEST	FINAL INSTALLATION TEST
1	Test of footballs	FIFA GLT TEST 01	-	Laboratory	-	FIFA FIFA QUALITY PRO / FIFA QUALITY or IMS standard	√	-
2	Test of goals	FIFA GLT TEST 02	Safety	Laboratory & field	-	BS EN 748:2013 + A1:2018 and additional evidence by an independent test institute	√	-
			Functionality			BS EN 748:2013 + A1:2018		
			Influence			Evidence by an independent test institute		
3	Test of durability of referee's watch	FIFA GLT TEST 03	Climatic	Laboratory	-	EN 60068	√	-
			Mechanical			Functionality after tests		
			Electromagnetic			EN 61000-4-2		
			Durability		50	>5h		
4	Laboratory test of critical components	FIFA GLT TEST 04	Defined by the test institute	Laboratory	-	-	√	-
5	Unrestricted shots	FIFA GLT TEST 05	-	Field	55	Correct indication of all shots during two sets of tests	√	√
6	Tests with target wall	FIFA GLT TEST 06	Free view	Field	42	Accuracy ± 1.5cm during two sets of tests	√	√
			Restricted view	Field	10	Accuracy ± 1.5cm for all shots	√	√
7	Slider test	FIFA GLT TEST 07	-	Field	20	Accuracy ± 1.5cm during two sets of tests	√	√
8	Tests of functionality of referee's watch	FIFA GLT TEST 08	Coverage	Field	10	Full pitch with run-offs during two sets of tests	√	√
9	Additional randomised shots on empty goals	FIFA GLT TEST 09	-	Field	25	Correct indication of all shots during two sets of tests	√	√
10	Training session	FIFA GLT TEST 12	-	Field	40	Correct indication of all shots	√	-
11	Tests of immunity to adverse conditions	FIFA GLT TEST 11	-	Field	50	ADDITIONAL INFORMATION ONLY	√	-
12	Goalkeeper test	FIFA GLT TEST 10		Field	30	Correct indication of all shots	√	√
13	Delay of referee's watch	FIFA GLT TEST 13	Goal indication		27	≤1second during two sets of tests	√	√
14	Product identification	-	-	In the stadium	-	Same component type as tested in FIF GLT TEST 04	-	√

Table 18: Overview of all test methods and requirements

5 TEST METHODS

This section describes the methods to be applied for testing.

5.1 Laboratory test methods

In order to determine that all components used in the GLT system are suitable for the application and to ensure reproducibility, an assessment of the GLT system is to be carried out in the laboratory. This series of tests is to be carried out independently from the field tests and can therefore be done before or parallel to the field tests.

5.1.1 Test of footballs – FIFA GLT TEST 01

Any type of ball used for a GLT system must comply with the FIFA Quality Programme for Footballs. It must meet the standard "FIFA FIFA QUALITY PRO", "FIFA FIFA QUALITY" or "IMS International Matchball Standard" (see section 4.1.1.1).

If the balls are not yet certified, this test must be carried out before a licence can be obtained.

Information regarding the testing scheme can be found on www.FIFA.com/Quality.

5.1.2 Test of goals (i.e. the goal itself) – FIFA GLT TEST 02

Goals that are modified by the provider of the GLT system have to comply with the Laws of the Game (Law 1). If changes to the shape of the goal are required to install/use the GLT system, the modified goal has to pass the following test.

5.1.2.1 Principle

A modified goal has to be tested by an independent test institute that is capable of carrying out the BS EN 748:2013 + A1:2018 test, according to the highest requirements for the safety of the players and functionality, and it must be proven that the modification has no influence on the game itself.

5.1.2.2 Apparatus

The goal has to be tested according to the BS EN 748:2013 + A1:2018 standard. At the request of the FIFA-accredited test institute, additional studies have to be carried out by independent test institutes in order to confirm that the shape of the goal is safe for players and has no influence on the game.

5.1.2.3 Procedure

The provider of the GLT system has to contact an independent test institute. The results of this test have to be submitted to FIFA by the FIFA-accredited test institute. Depending on these results, FIFA decides if further studies by independent test institutes are necessary (see section 4.1.1.2).

5.1.3 Test of durability of referee's watch – FIFA GLT TEST 03

The transmission of the goal signal from the main computer of the GLT system to the referee's watch is an essential component of the GLT system. As the referee's watch is a critical component of each GLT system, a separate testing protocol shall ensure that the technology works reliably. Therefore, qualification tests in the laboratory are performed with test parameters in accordance with IEC 60068 (Environmental testing) and 61000-4-2 (Electromagnetic compatibility). These comprise:

- vibration and shock testing
- temperature and humidity testing
- immunity to electrostatic discharge

For the laboratory qualification test, at least three referee watches must be made available together with a wireless activation device (sender) and all necessary instructions to verify

the correct functioning of the watch (see section 4.1.1.3). In order to pass the tests described in section 5.1.3 (tests 1-4), all tested watches have to show the same functionality as they displayed before the test.

5.1.3.1 Vibration and shock testing

5.1.3.1.1 Principle
Vibration and shock testing are used to evaluate the mechanical robustness of the referee's watch, which is exposed to mechanical stress in use, transport and storage. In vibration tests the watch is exposed to increasing frequency but constant acceleration. The shock test applies a high acceleration, while the free fall test is a real life test with less defined boundary conditions.

5.1.3.1.2 Apparatus
A shaker with a frequency range of at least 10-1000Hz and acceleration of at least 20g is required. A three-axis mounting device is needed to mount the watch. The load-bearing capacity of the shaker must conform with the weight of the referee's watch and the three-axis mounting device.

5.1.3.1.3 Procedure
The x-y-plane is defined by the cover glass, the x-axis being aligned parallel to the straps; the z-axis perpendicular to the cover glass.

The watch is switched on and its functionality is checked. The casing is fixed to the mounting device in a first orientation and the straps are secured. The mounting device is fixed to the shaker and a sine sweep from 10 – 500 Hz at 1 octave per minute with a constant acceleration of 2g is applied for 1 hour. Then, mechanical shock with an acceleration of 15g, 11ms half sine, is applied and repeated 10 times.

The procedure is repeated for the other two orientations. After each orientation, the functionality of the watch is checked.

Finally, the watch is dropped onto a concrete floor from a height of 1m, 10 times in sequence. At the end, the functionality of the watch is checked.

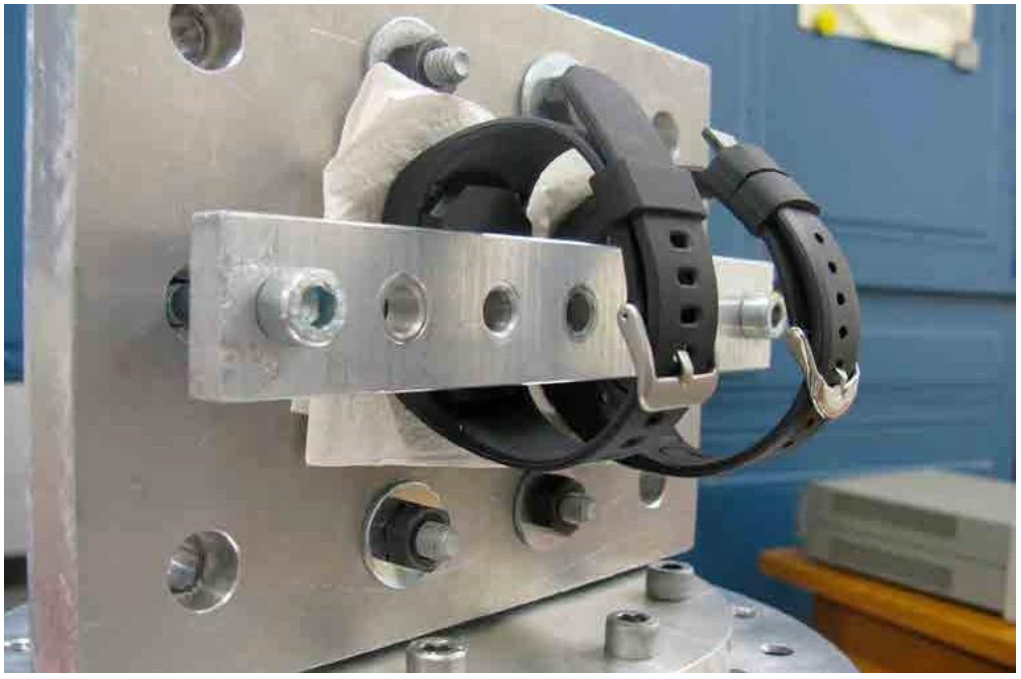


Figure 2: Mounting of two referee's watches on a shaker. Test parallel to x-axis.

5.1.3.2 Temperature and humidity testing

5.1.3.2.1 Principle
Temperature and humidity testing are used to evaluate the environmental robustness of the referee's watch, which is exposed to environmental stress in use, transport and storage. High and low temperatures combined with elevated humidity are challenging environments for electronic equipment.

5.1.3.2.2 Apparatus
A climatic chamber with a temperature range of at least -20°C – +60°C and humidity levels of up to 95% r.H. is required. The window of the climatic chamber must allow for observation of the goal indication during the test.

5.1.3.2.3 Procedure
The watch is switched on and mounted in the climatic chamber such that the goal indication can be read through the window.

1) Low temperature test: The temperature is lowered to 0°C for 2 hours and the functionality is checked while in the chamber. The test is repeated for -10°C and -20°C.
2) High temperature test, dry conditions: The temperature is raised to 40°C for 2 hours and the functionality is checked while in the chamber. The test is repeated for 50°C and 60°C.
3) Humidity test: The temperature is set to 40°C and the humidity to 93% r.H. The watch is kept in the chamber for 2 hours and the functionality is checked while in the chamber.
4) Climatic cycle: The watch is switched off and mounted in the chamber. Two temperature cycles from 25°C to 40°C and back at a constant 93% r.H. are performed, followed by a cycle to -10°C and back to 25°C. Total time in the chamber is 24 hours. After the cycle, the watch is taken out and its functionality is checked.
5) Immersion test: Finally, the watch is switched off and immersed in 15cm of water for half an hour. After removing and drying the watch externally, its functionality is checked.

5.1.3.3 Electrostatic discharge testing

5.1.3.3.1 Principle
Electrostatic discharge (ESD) can happen every time the watch is in contact with a person. This test simulates ESD with defined parameters.

5.1.3.4.2 Apparatus
An ESD gun with at least 2kV discharge voltage is required.

5.1.3.4.3 Procedure
The watch is switched on and an ESD contact discharge of 2000V is fired to a metallic part of the watch. The functionality is checked. The test is repeated for all other metallic parts and the display of the watch.

5.1.3.4 Durability check

5.1.3.4.1 Principle
During the tests, the referee's watch must be operational for at least 5 hours without interruption. This is to be measured during the series of tests.

5.1.3.4.2 Apparatus
Three referee's watches, as provided by the GLT provider, are tested in laboratory conditions.

5.1.3.4.3 Procedure
The watches are switched on and placed in laboratory conditions. They are activated with a goal signal transmission at regular intervals of 5 minutes. Correct goal indication is checked after 5 hours.

5.1.4 Laboratory test of critical components – FIFA GLT TEST 04

5.1.4.1 Principle
Depending on the nature of the GLT system, different components will need testing for their functionality. The principle of this test is to identify and qualify all components that are necessary for the operation of the GLT system (see section 4.1.14). The aim of this section is to prevent technology that has obvious flaws that are not taken into account by the current test methods. A costed testing proposal should be recommended to FIFA by the test institute and subject to FIFA's approval, the testing can take place at a cost to the provider.

5.1.4.2 Apparatus
The suitable apparatus for testing the components of the GLT system will be determined by FIFA and the FIFA-accredited test institute following, wherever possible, established international standards for evaluation.

5.1.4.3 Procedure
The GLT provider must provide the FIFA-accredited test institute with all components of the GLT system as well as a system sheet exactly defining these components. Besides being the benchmark of the laboratory test, this system sheet will be the minimum requirement for any installation of that particular GLT system. The following procedure has to be conducted:

a) Submission of all components of the GLT system to the FIFA-accredited test institute
b) Identification of critical components by the FIFA-accredited test institute
c) Definition of tests for the identified critical components by the FIFA-accredited test institutes
d) FIFA's written approval is mandatory for the execution of these defined tests
e) All tests that are defined by the FIFA-accredited test institute must be approved by FIFA

All identified and critical components of the GLT system will be tested according to a test methodology intended to determine the functionality of the component in question.

Typical tests will include the following elements:
– vibration and shock testing
– environmental testing (humidity and temperature)
– immunity to electric distortions (electrostatic discharge, electromagnetic compatibility)
– additional specific tests for the transmission technology used by the GLT system

5.2 Field test methods
This section describes the principle and procedure of each of the FIFA GLT tests that are performed on the field as well as indicating the apparatus needed for each test. More details are found in the GLT system test protocol (see table 10 for details of the requirements). When reference is made to installing test equipment at precise angles, a digital protractor, water level or plumb-line with an accuracy of ±0.1° is to be used. For HD video camera tests, a minimum of 6 hours of recording time must be ensured at a rate of at least 30 frames per second. One camera is positioned on the goal-line with a stopwatch continuously running and the other is placed on the corner of the 6-yard box. Both cameras shall have a clear view of a referee's watch.

5.2.1 Unrestricted shots – FIFA GLT TEST 05

5.2.1.1 Principle

This scenario provides overall information on whether the GLT system would reliably detect a goal and would be able to distinguish between a goal and a no-goal situation. The scenario is based on an empty goal free from any measurement equipment and hence with an unrestricted view of the ball. It is, however, possible for one technician to stand in the goal for collection of the unrestricted shots to reduce wear on the surface.

5.2.1.2 Apparatus

The apparatus to be used for carrying out shots is a ball shooting machine. Different ball shooting machines may be used with the following minimum specifications:

- Ball speed at exit: 20-100 km/h
- Axis of spin: 0-360 degrees about the direction of flight

Ball speed has to be adjustable to 35 km/h, 50km/h and 70 km/h (± 2.5 km/h). Examples of ball shooting machines that fulfil these requirements are “SideKick” and “Globus” (see Figure 3). A stop watch is used to measure the goal indication delay. (Seattle Sport Sciences, Inc., 24066 NE 53rd Pl.; Redmond WA, 98053; USA).





Figure 3: “SideKick” and “Globus” ball shooting machines

5.2.1.3 Procedure

The ball shooting machine is positioned at a distance of 7 metres (± 0.5) from the goal (measured from the back of the goal line) and 2 metres (±0.25) to the left of the right post (see Figure 4). Should the test institute select to position the target wall closer to the left post, the ball shooting machine is set up accordingly on the other side of the goal. For post rebound shots, the ball shooting machine is placed at a distances of 7 metres (± 0.5) from the goal and aligned with either of the goalposts.

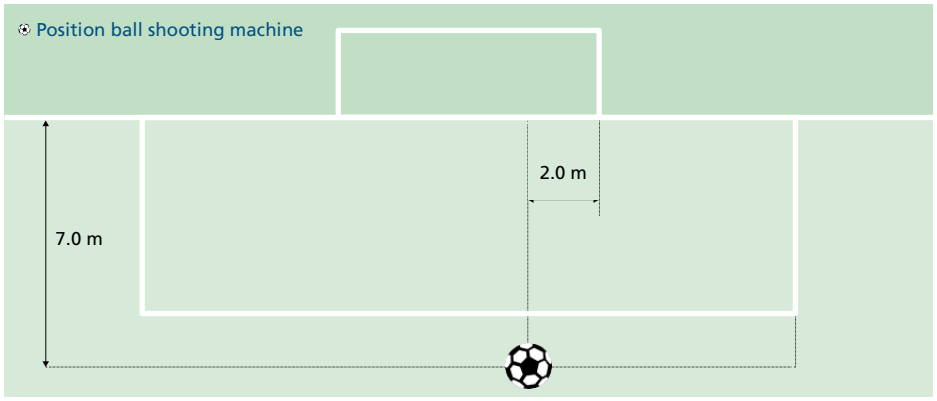


Figure 4: Position of ball shooting machine

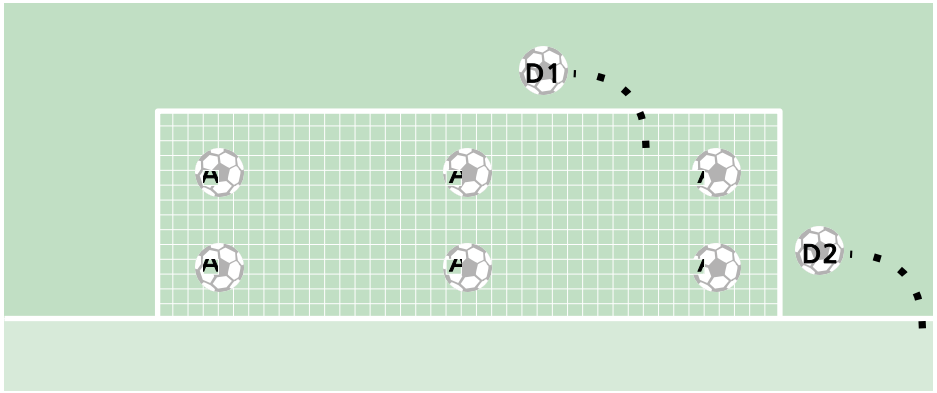


Figure 5: Examples of free shots

For this test, the shots to positions A01-A06 balls shall be shot with the ball shooting machine as indicated in Figure 5. Shots to the positions D01-D02 shall be performed manually by one of the testers. Shots in D01 must be thrown over the crossbar and D02 shall be kicked into the outer netting at varying speeds. The position of each shot is at the discretion of the test institute. Five shots are performed for each of these positions. Indirect shots A07-A08 are performed with a player standing on either side of the 6 yard box and passing the ball into the goal area for the player to shoot. The post rebound shots A09 are performed with the ball shooting machine placed at 7 metres (+/- 0.5m) from the goal aligned with either of the goalposts.

5.2.1.4 Calculation and expression of results

A total of 55 shots are performed and used as the basis for calculations. All results will be recorded by the responsible tester and at least a second person, noting whether the goal, near miss or shot in the outer netting was correctly recognised (Goal or No Goal). In case of a long delay for goal indication on the referee's watch, the exact delay is evaluated using the footage from the HD camera on the goal-line.

5.2.2 Tests with target wall – FIFA GLT TEST 06

5.2.2.1 Principle

One of the critical test scenarios for evaluation of the GLT system is the target wall test. The target wall simulates the goalkeeper or a player inside the goal area. With the target wall standing behind the goal line at a defined distance, the ability of the GLT systems to detect goals when the ball crosses the line for only a fraction of a second is assessed. In this way, temporal resolution can also be assessed. Since the ball is only a few centimetres behind the line, geometric accuracy is as crucial as the time resolution.

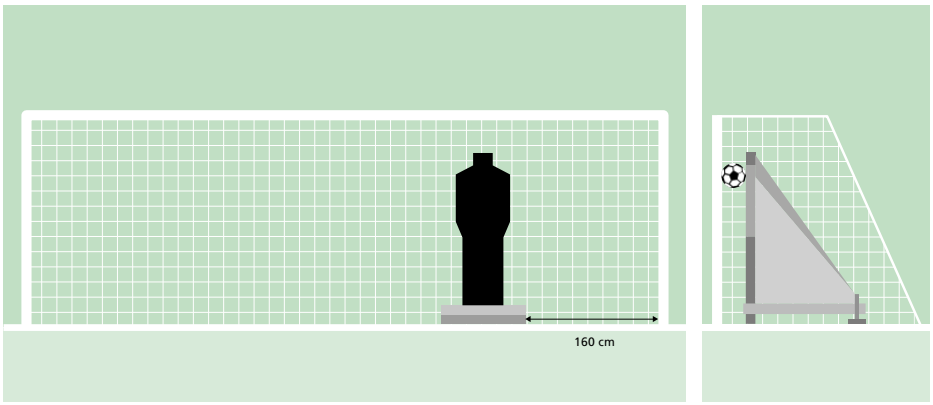


Figure 6: Scenario 2: Shots on the target wall

5.2.2.2 Apparatus

The target wall should be built entirely of non-metallic materials with the exception of a few short screws. The wall is positioned manually to prevent damage to the pitch and is loaded with dead weight (e.g. water tanks) to avoid a relocation of the position from shot to shot. It has roughly the shape of a player.



Figure 7: Target walls used GLT tests

Technical data

- Dimensions: Base plate: 80 x 120cm; Target wall: height 192 ±10cm; width: 70 ±5cm at the widest part, 50 ±5cm at the base and 30 ±5cm at the top. The widest width shall be at the height starting at 65 ±10cm and ending at 150±10cm. Refer to CAD drawings in Appendix A for example.
- Material: Rigid surface with a maximum deformation of 5mm using the 70km/h impact of the ball shooting machine, plastic screws for levelling
- Positioning: Front edge aligned parallel to goal line (line remains fully visible)
- Fixation: Dead weight (e.g. water tanks)
- Weight: The total weight (including dead weight) is at least 220kg

CAD drawings for detailed technical information to be used as reference are in Appendix A.

The dummy players to be used for the tests with restricted view are to be made entirely from non-metallic materials to prevent disturbance of systems measuring electromagnetic fields and have the size and shape of an average football player (minimum dimensions, height: 165cm, width: 50cm, shoulder height 140cm). The dummies shall be dressed with a football kit (football shirt, shorts and football socks). Two different colours (two shirts in one colour and two in another) shall be used for the shirts.

A high-speed camera is to be used for parts of the tests. The definition of the high-speed camera must be no worse than 1.5mm per pixel for the recorded area between the target wall and the goal-line. Because of the short exposure time, intense lighting could be needed, especially under floodlight conditions. Therefore, for some of the tests, a flicker-free lamp can be used. The camera should be placed inside the goal whenever possible and as far away from the target wall as possible. A protective shield may be used if required. If the flicker-free lamp is necessary it is positioned inside the goal area to illuminate the ball during the impact on the wall.

The pressure of the ball shall be measured before and after the set of target wall tests to ensure there was no significant change.

5223 Procedure

5.2.2.3.1 Tests with unrestricted view

The front part of the target wall is aligned parallel to the goal line and be vertical (90 ±0.2°) inside the goal. In order not to disturb camera-based systems, the goal line has to be fully visible at all times. The target wall should be placed at a distance of 160cm inside the right post when facing the goal. The test institute may decide to place the target wall on the inside of the left post at its discretion. All tests within one stadium shall be carried out with the target wall in the same position. When the target wall is placed on the left, the ball shooter shall be moved accordingly to be aligned with it. All shots B01-B07 are to be carried out with a shooter speed of 50km/h (±2.5km/h)

For the system test, shots B01-B07 are carried out (see section 4.1.2.2). Since the ball is deformed when hitting the wall, it is also possible to verify if the system correctly accounts for this effect. A high-speed camera is used for part of these tests in order to verify the goal event in function of the distance of the target wall. This is done by detailed analysis of the slow and fast shots (different ball speeds will result
in different deformation intensities and slightly different trigger points of goal situations). All shots should hit the target wall between a height of 20 and 165cm. The distance of the target wall to the goal-line and the verticality shall be measured before and after each series of shots at a specific distance. Should there be a variation of ±2mm in distance or verticality outside the specified range of 90° ±0.2, the block shall be repeated.

For the final installation test, the shots B01 are carried out and the footage from the high-speed camera is used to determine the position of the target wall for which the ball has crossed the line entirely. Once this measurement has been established, shots B02-B03 are carried out and must give clear "goal" and "no goal" indications (see section 4.2.2.2). Since the ball is deformed when hitting the wall, it is also possible to verify if the system correctly accounts for this effect. A high- speed camera is used for part of these tests in order to verify the goal event in function of the distance of the target wall. This is done by detailed analysis of the slow and fast shots (different ball speeds will result in different deformation intensities and slightly different trigger points of goal situations). All shots should hit the target wall between a height of 20 and 165cm. The distance of the target wall to the goal-line and the verticality shall be measured before and after each series of shots at a specific distance. Should there be a variation of ±2mm in distance or verticality outside the specified range of 90° ±0.2, the block shall be repeated.

During the tests, the camera is positioned inside the goal, and the centre of the image is carefully aligned with the goal line using two reference plumb lines, which are positioned next to each side of the target wall.
Alternatively the camera may be aligned onto the goal line using a precision geo-system line laser which is attached along the centre line of camera lens axis. It must have vertical and horizontal line capacity and have a self-levelling functionality. The laser specification is 635nm - Class II. Minimum Vertical line and dot laser function.
The accuracy of measurements should be:
- Levelling accuracy @ 5 m ± 1.5 mm
- Accuracy of plumb point @ 5 m ± 1.5 mm
- Accuracy of horizontal line @ 5 m ± 1.5 mm
- Vertical accuracy @ 3 m line length ± 0.75 mm



Figure 8: a) No goal and b) goal images captured at 2000 fps. The red line schematically shows the goal line.

Figure 8: a) shows the ball at its maximum deformation during a no-goal shot (wall at 120mm, ball speed of 35km/h) and a goal shot (240mm, 35km/h). A line can be added digitally to serve as a visual indication of the goal line on the level of the impact wall.

The ball is shot onto the target wall using the ball shooting machine as set up in FIFA GLT TEST 05.

For the system test, a total of 94 shots are performed and used as the basis for calculations. At least 24 shots must be clear goal situations (≥240mm distance to the goal line). For shots B01 the delay of the goal indication is evaluated using the footage from the HD camera positioned on the goal- line.

For the final installation test, a total of 36 shots are performed and used as the basis for evaluation. For shots B02 the delay of the goal indication is evaluated using the footage from the HD camera positioned on the goal- line.

5.2.2.3.2 Tests with restricted view
For this test, two dummy players are positioned on each side (total of 4 dummy players) to partially block the view before the ball hits the target wall. This scenario is selected to simulate the situation when players and/or the goalkeeper prevent a direct view of the ball. Camera-based systems will have to make sure that, by using multiple cameras and/or optimised positioning and viewing angle of the cameras, the ball will be always visible.

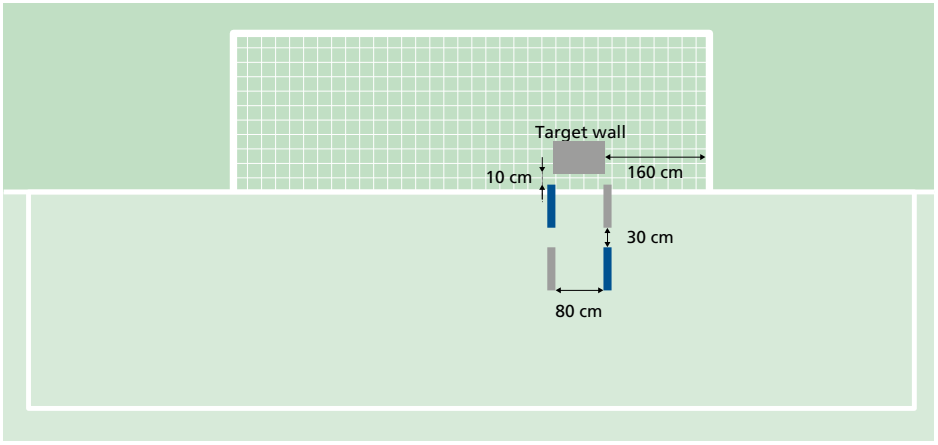


Figure 9: Positioning of four dummies

This set-up applies to the target wall. Two dummies are each centred at a right angle to the goal line at a distance of 80cm (±5cm) apart. This distance shortest distance between the dummy player and the target wall shall not exceed 10cm. A second series is in line with the first row with a free distance of 30cm (±2.5cm) at the closest point. The target wall will be located in the middle between the dummies. Shots C01 shall be carried out at 35km/h (±2.5km/h) and shots C02 at 70km/h (±2.5km/h)

For this test, 5 shots of each scenario C01 and C03 (see section 4.1.2.2 and 4.2.1.2) are performed and used as the basis for calculation. If a shot hits the target wall and rebounds back clearing both rows of dummies, it is valid irrespective of any contact that may have occurred with a dummy player. In case the ball does not rebound back beyond the dummy players or at least one of the dummy players falls over, the shot shall be repeated.

5.2.2.3.3 Tests with indirect shots and restricted view
In order to have as realistic a simulation as possible, a set of indirect shots are performed with the dummies acting as obstacles. For this test, 4 dummies are used.

For this test, 5 shots of each test scenario C02 and C04 (see section 4.1.2.2 and 4.2.1.2) are performed and used as the basis for calculation. If a shot hits the target wall and rebounds back clearing both rows of dummies, it is valid irrespective of any contact that may have occurred with a dummy player. In case the ball does not rebound back beyond the dummy players or at least one of the dummy players falls over, the shot shall be repeated.

5.2.3 Slider test – FIFA GLT TEST 07

5.2.3.1 Principle
In order to determine the exact point at which the GLT system triggers and indicates the goal, the slider test device is placed on the goal line and the ball is slowly and progressively moved across the line. A precise measurement of the location of the ball when the system is triggered is possible in this test.

5.2.3.2 Apparatus

A slider or translation stage is used to move the ball slowly (approx. 0.5cm/s) across the goal line in order to determine the exact trigger point for goal detection. The ball should be placed in a way that it can spin freely when it is being moved on the slider.

Technical data

- Dimensions: Basis: min 60 x 40cm
- Material: non-metallic, means for levelling
- Fixation: No additional fixation

An example of CAD drawings for detailed technical information is in Appendix B.





Figure 10: Different sliders used to assess the trigger point for goal detection

5.2.3.3 Procedure

The frame of the slider is positioned over the goal line so that the centre of the ball is positioned at 0.8m (±0.05m) from the goal post (see Figure 11). The slider test is performed on the side of the goal on which the target wall tests were not performed. The same side is to be used for the test on both goals. With the ball on the bars, the position of the goal event is assessed using a plumb line or equivalent technique (orthogonal to the pitch and the inner edge of the goal line). To start the measurement, the ball is taken from a position outside the 6-yard box and placed on the outer end of the slider. The operator is working from the goal mouth and moves the ball continuously such that it rolls across the goal line until the GLT system indicates a goal. The position of its back end with respect to the previously found zero position of the goal line is recorded and allows the offset where a goal is triggered to be determined. The test is repeated with the slide positioned in the centre of the goal at 3.6m (± 0.05m).

A total of 10 goal events in two different positions E01/E03 and E02/E04 are performed and used as the basis for calculations (see section 4.1.2.4). This test is to be performed on both goals.

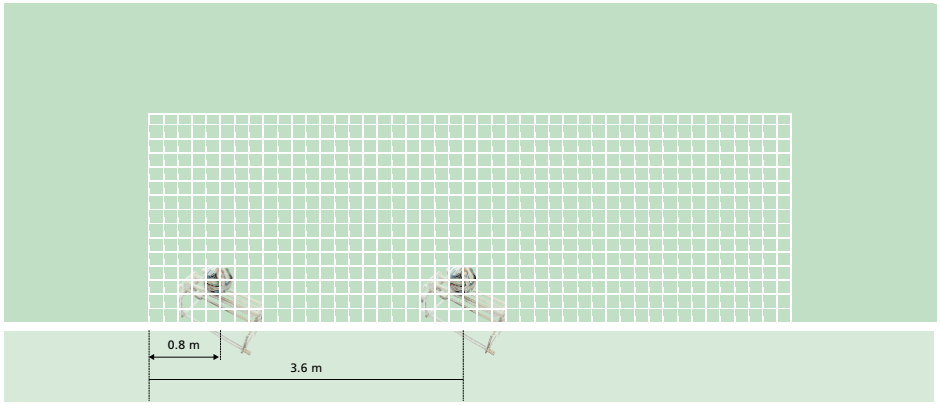


Figure 11: Position of slide on the goal line

5.2.4 Tests of functionality of referee's watch – FIFA GLT TEST 08

5.2.4.1 Principle

In order to ensure that the referee would get the indication of a goal anywhere on the pitch and within the required time of one second, the watch is tested for these parameters.

5.2.4.2 Apparatus

No apparatus other than a stopwatch, six referee's watches and an interfering transmitter (see section 5.2.7.2) is needed for this test.

5.2.4.3 Procedure

The functioning of the referee's indication device is tested according to the following specifications.

5.2.4.3.1 Indication to referee only

To ensure that only the referee receives the indication through visual confirmation on the watch and vibration, it must be assessed that no other form of confirmation is emitted.

5.2.4.3.2 Range check; full coverage of goal indication on referee's watch
Goal indication has to work in all the positions on the field a referee or assistant referee might be in during a match (see section 4.1.2.4). Two members of the test team position themselves at the corner points of the goal. One of the two people is to wear two referee watches; a third member is on the penalty point with one watch; the person responsible for the test remains in the goal area with the ball, one watch and an interfering transmitter. The person responsible for the test throws the ball into the goal and all watch indications are recorded by all testers (the ball shooting technician checks the watch that is setup in front of the HD cameras).
Then, the testers at the corners and the penalty point move to the centre line (see Figure 12). The person responsible for the test throws the ball into the goal again and all watch indications are recorded by all testers. Then the testers move to the opposite goal and corner and the procedure is repeated. On the way back, the procedure is repeated in at least two positions, including the area which is at least five (5) meters outside the perimeter of the pitch. This test is then repeated with the person throwing the ball into the goal activating the interfering transmitter for all positions. Should the transmitter allow for the two series to be carried out simultaneously, the test institute may perform two shots per position – one without transmitter and one with.

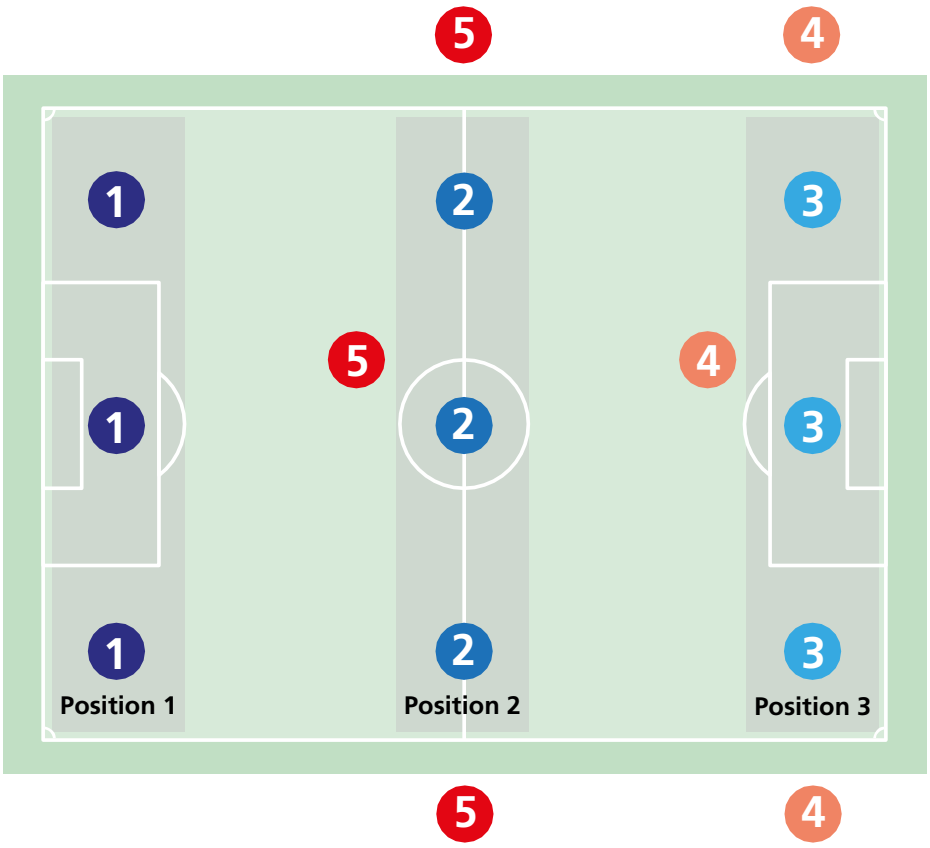


Figure 12: Full coverage of goal indication on referee's watch

5.2.5 Additional randomised shots – FIFA GLT TEST 09

5.2.5.1 Principle

In addition to the standardised tests, each GLT system will be tested with a series of shots into the goal that the testers carry out at random during the pre-defined testing blocks to check whether the GLT system is continuously operative. The random shots consist of two different types: shots with the ball that must indicate a clear goal and shots with non-ball projectiles that must indicate a "no-goal".

5.2.5.2 Apparatus

The testing ball is to be used for the ball shots. Non-ball projectiles are objects that would typically be used during a football match and have similarities to a football. These shall be goalkeeper caps, goalkeeper gloves, towels, water bottles and football boots in a colour similar to the ball. Each projectile should be used on at least 3 occasions.

5.2.5.3 Procedure

During the testing blocks (see section 4.1.2.5 and 4.2.1.5), the tester shoots the ball or throws the non-ball projectile into the net from random positions, at random speed. For all tests, note is taken of whether a goal was indicated.

5.2.6 Goalkeeper test – FIFA GLT TEST 10

5.2.6.1 Principle

Shots on a goalkeeper are used to simulate real situations. The situations include catching the ball with both hands in front of the body as well as catching the ball and embracing it with both arms. The latter version covers a considerable part of the ball, increasing the chance that ball tracking will be lost by camera-based systems. In the third situation, the ball rolls into the goal beneath the goalkeeper.

5.2.6.2 Apparatus

For the goalkeeper test, a human tester dressed in goalkeeping gear (shirt and gloves) is used. While size and weight are not critical, they should ideally be representative of a goalkeeper.

5.2.6.3 Procedure

A) The goalkeeper should be positioned approximately 1m in front of the goal line. The ball is delivered at moderate speed either by the ball shooting machine or a player. The goalkeeper catches the ball by placing both hands on the side of the ball in front of the body at shoulder height with outstretched arms (see Figure 13). Holding the ball in this position, the goalkeeper slowly walks backwards over the goal line (approx. 0.5 m/s) until the ball is at least 20cm behind the goal line, then returns to the field. This scenario F01 is repeated five times with the ball and five times without the ball. The test institute determines the order of shots and may alternate between shot with and without the ball at its discretion. When conducting each test without the ball the goalkeeper should punch the ball and clear the thrower before their movement back into the goal

B) The goalkeeper should be positioned approximately 1m in front of the goal line. The ball is delivered at moderate speed either by the ball shooting machine or a player. The goalkeeper catches the ball standing in an upright position and embraces it with both arms and forearms parallel holding it against their chest (see Figure 14). Holding the ball in this position, the goalkeeper slowly walks backwards over the goal line (approx. 0.5 m/s) until the ball is at least 20 cm behind the goal line, then returns to the field. This scenario F02 is repeated five
times with the ball and five times without the ball. The test institute determines the order of shots and may alternate between shot with and without the ball at its discretion. When conducting each test without the ball the goalkeeper should punch the ball and clear the thrower before their movement back into the goal.

C) The goalkeeper should be positioned with their feet on the goal line. They crouch down and form a tunnel with their arms and legs, fingertips touching the ground. The ball is placed 1m in front of the goalkeeper, and a player hits it gently so that it rolls through the tunnel into the goal (see Figure 15). The position of the goalkeeper is left to the test institutes' discretion. At least one shot is to be performed close to either goal. This scenario F03 is repeated five times with the ball and five times without the ball. In the fourth and fifth repetition of both cases one player moves into the goal and kicks the ball out. The test institute determines

the order of shots and may alternate between shot with and without the ball at its discretion.

For all tests (see section 4.2.1.6), note is taken of whether a goal was indicated. If so, it must be verified that the whole ball circumference was over the line at the time of goal indication and the delay of indication must be estimated.

Evaluation:
A total of 30 shots are performed during the goalkeeper test. 10 shots are performed for each version of which 5 are with and 5 without the ball. The sequence of shots (with or without ball) is left to the discretion of the test institute. Note on the protocol sheet whether a goal was indicated, verify if the whole circumference was over the line at the time of goal indication and also check delay of indication.

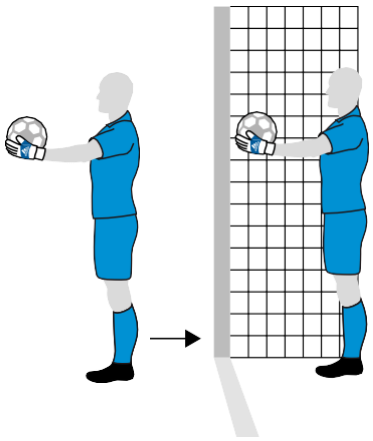


Figure 13: A) Catching the ball with the hands in front of the body

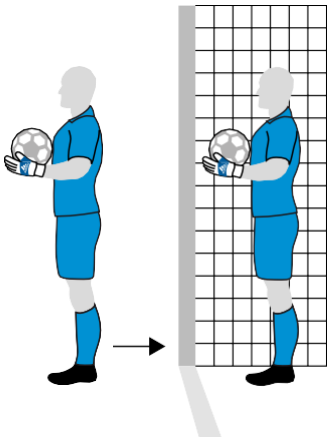


Figure 14: B) Catching the ball with arms and embracing it

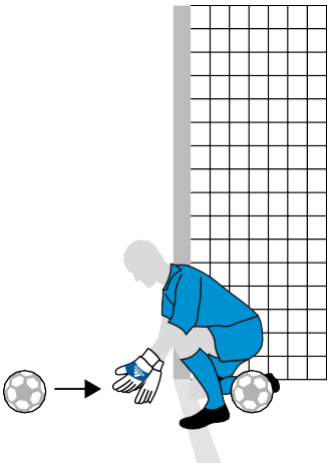


Figure 15: C) Crouching position, ball rolls into goal

5.2.7 Test of immunity to adverse conditions – FIFA GLT Test 11

5.2.7.1 Principle

Several tests are performed to assess the immunity of a GLT system to adverse conditions. Simulated situations on the field and its environment include the presence of objects and materials that players or fans might bring or throw on, sudden climatic changes, wear-out of the ball, or faulty electrical installations. Specific equipment is used for these tests. Some of the tests may influence all potential GLT systems, whereas others are designed to test specific classes of systems (such as camera-based systems or systems using electromagnetic fields).

5.2.7.2 Apparatus

These tests require a set of specialised material and equipment which may depend on the GLT system.
For all GLT systems the following materials are required:

- Bright objects (e.g. white toilet paper rolls)
- Metallic objects (e.g. metallic water bottle, min. 0.33l; metallic wrench, min. 0.3kg)
- Two sheets of white linen (min. 1.2m wide and 7m long; min. 1.2m wide and 10m long)
- Smoke machine based on glycol/water fluid with a hose of at least 5m in length
- Mobile phone
- Wireless garage opener or other activating device as interfering transmitter (ISM frequency band around 863-870 MHz, rated operating distance at least 100m (free field))
- Goalkeeper cap, goalkeeper gloves and towel in white colour

For camera-based systems, the following are also required:

- Flashlights (professional camera flashlight or high-speed camera illumination strobe)
- Laser pointers (two green laser pointers, wavelength around 532nm, min. 5mW power)
- Balls with additional or worn out labels/marks (three places, dark colour, at least 0.5 x 2 each)

For magnetic field-based systems, the following are also required:

- Single wire power lines (length at least 10m) with two junctions to separate and recombine phase and neutral conductor of the power cord
- Electric device capable of burst discharges or generating electromagnetic fields by switching on a load of at least 500W

For RFID-based systems, the following are also required:

- Radio frequency source, adapted to the frequency band used by the GLT system (e.g. Bluetooth sender in the 2450 MHz band)

5.2.7.3 Procedure

A set of tests G01-G09 will be performed under adverse conditions.

1) **Presence of bright objects (G01):** Place the metallic water bottle inside the goal halfway between the goal line and the corner of the net. Throw two toilet paper rolls from the back on top of the goal (centre and left side), allowing it to unroll; place a third toilet paper roll next to the goal (right side), one metre from the base line and touching the net. Perform five free shots with low speed according to the protocol for unrestricted shots.

2) **Snow simulation test (G02):** Unroll the short linen inside the goal parallel to the goal line; unroll the long linen outside the goal parallel to the goal line. Leave a minimum of 15cm space to the goal line. Wet the ball. Perform five free shots with low speed according to the protocol for unrestricted shots.

3) **Dirty ball test (G03):** Perform five free shots with low speed according to the respective protocol using the ball with additional marks which should simulate a dirty ball.

4) **a) Laser pointer test (G04):** (camera-based systems only) The regulations and instructions for laser safety have to be followed (see A4). Target two GLT cameras on the same side of the stadium with the two lasers. Activate lasers during the shots. Perform five free shots with low speed according to the protocol for unrestricted shots.

   b) Field distortion test (G04): (magnetic field-based GLT systems only) Mount the metallic wrench on a cord and let it hang from the goal bar, 2m from the left post. Length of the cord is at least 1m. Let the wrench swing during the shots. Perform five free shots with low speed according to the protocol for unrestricted shots.

5) **Flashlight test (G05):** Activate the flashlight (at least three strokes within two seconds) or strobe the illumination during the shots. Perform five free shots with low speed according to the protocol for unrestricted shots.

6) **Smoke test (G06):** Use the smoke generator to simulate the presence of fog or use of a smoke petard. Place the generator upwind and use the hose to create fog on top of the goal area. Wait until the goal is sufficiently visible (that a match would be possible) and perform five free shots with low speed according to the protocol for unrestricted shots.

7) **Interfering transmitter test (G07):** Activate the mobile phone, wireless garage opener, referee communication system or other activating device near the referee watches and note any goal indication of the watches. Repeat for all transmitter channels (buttons). Then, activate the devices during the shots. Perform five free shots with low speed according to the protocol for unrestricted shots.

8) **Second ball interference (G08):** Place a second ball outside the goal, 1m behind the goal line, 1m from the net. Perform five free shots with low speed according to the protocol for unrestricted shots. Remove and reposition the second ball before each shot.

9) **Line test (G09):** The single wire power line (phase conductor) is placed behind the goal next to the net with the electric device (load) on one side and power connection on the other side. The neutral conductor is laid at a distance of at least 2m parallel to the power line. The electric device is activated or switched on during the shots. Perform five shots with low speed according to the protocol for unrestricted shots.

10) **Non-ball projectiles (G10):** Projectiles that would typically be found on a football pitch shall be thrown into the goal the same way the ball is. Such projectiles are goalkeeper caps, goalkeeper gloves, towels, water bottles and football boots. The objects should have a colour similar to the ball that is being tested.

A total of 50 shots are performed for all GLT systems to assess immunity to adverse conditions. Five shots per type of inference are to be used for calculation of results.

5.2.8 Training session – FIFA GLT TEST 12

5.2.8.1 Principle

To simulate goal situations with some degree of unpredictability, a training session will be arranged by a local football club. The provider should lead the organisation of this test; responsibilities include the organisation of 14 players, changing rooms, full kits (7 home and 7 away) and the liaison with ground staff. The test institute is responsible for instructing the players for the test.

A total of four scenarios are played on both goals. Corner balls (left/right) and free kicks (9m/17m) into the penalty area are performed by at least 14 players in two different shirts corresponding to two teams of which one is to play the position of goalkeeper. This test setting simulates standard situations of a real game and implies many goal situations, some of which might be critical. The training sessions must be recorded with HD video cameras to have proof of goal events which were related to the goal indication from the referee's watch.

5.2.8.2 Apparatus

Fourteen players from a local football club dressed in home and away shirts (7 v. 7) will be needed for this test.

5.2.8.3 Procedure

The players simulate set pieces (see 4.1.3). The order and frequency of the events are clearly dictated by the testing team. Should one set piece not lead to a goal situation it should be repeated so as to have sufficient data. All players apart from the set piece taker are in or on the edge of the 6-yard box during the following tests.

40 shots are performed for this test. At least the following situations and number of repetitions will be applied for both goals:

2 x 5 corner situations (H01, H02, H05, H06) per side (see Figure 16)
2 x 5 free kicks from two different distances (H03, H04, H07, H08) per side (see Figure 17)

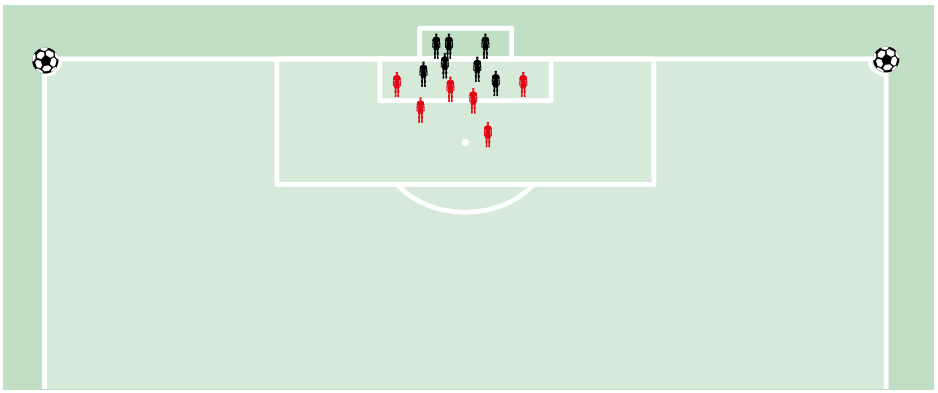


Figure 16: Positioning for corner kicks

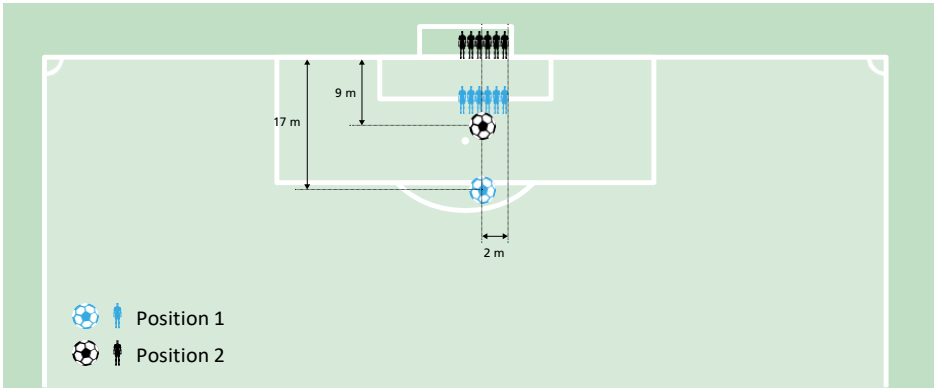


Figure 17: Positioning for free kicks

5.2.9 FIFA GLT TEST 13 – Goal signal indication delay of the referee's watch

This test is always performed in parallel to one of the other tests as the aim is to evaluate whether as well as giving the correct indication, the system is doing so in the required time. The signal on two watches must indicate the correct decision within one second of the ball crossing the goal line.

5.2.9.1 Principle
In order to determine the exact delay of the signal indication, video footage is used on which both the physical crossing of the line by the ball and two of the referee's watches are visible. A continuously running stopwatch is used to evaluate the delay.

5.2.9.2 Apparatus
The HD camera used to record the tests on the goal-line (outside the goal), a referee's watch and a stopwatch.

5.2.9.3 Procedure
For test blocks A01, A07, B02 and F01, the testers evaluate the footage from the HD camera that has recorded all 27 shots in which the ball crossed the goal-line. The image contains the direct view of one of the referee's watches, a continuously running stopwatch and a clear view of the goal-line. All 27 shots must be indicated correctly within 1 second (≤1s).

Any shot that is not included in the 27 above but that is part of the field test, may not exceed 5 seconds. In case a long delay occurs, the test institute shall use the footage from the HD camera on the goal-line to determine the exact duration of the delay. No individual shot may take longer than 5 seconds to correctly indicate a goal.

6 MAINTENANCE

Once the system has been tested, the challenge is to maintain its quality and functionality. Aftercare is an essential part of the sustained quality assurance of the GLT system, which is why the GLT provider must commit to performing regular post-installation maintenance. As part of the final installation test, the customer must confirm that the GLT provider has provided a maintenance manual, maintenance equipment and a maintenance training session for its GLT system to the customer.

7 WARRANTY

It is the GLT provider's obligation to provide its customer with a warranty of at least 2 years for each installed GLT system (including for all its components) at a given stadium. The length of the warranty has been chosen based on the existing standard in many countries. FIFA will only certify a given GLT system installation if it has been confirmed by the customer that it has been granted a 2-year (or longer) warranty.


FIFA Club World Cup Japan 2012
TOYOTA
VISA
VISA
Fly Emirates
Rigaos
SONY
SONY



CHECK BY THE REFEREE

The IFAB has clarified the referee's position in the Laws of the Game (Law 5) by stating that match officials can use the GLT system as a support provided they are convinced of its functionality, for which appropriate tests shall be carried out before the match.

1 TESTS BEFORE THE MATCH

The match officials are obliged to check the functionality of the GLT system by means of specific tests before it is used prior to each match. If the tests do not satisfy the referee, they can reject the use of the GLT system and therefore decide not to use it for the relevant match. One hour before kick-off, the stadium operator, club, competition organiser (as appropriate) or GLT provider has to hand over the match balls and 6 referees' watches
to the match officials for the final check and the decision on the use of the installed GLT system.

Once the decision has been taken as to whether the GLT system will be used or not in a match, the stadium operator must be immediately informed. Furthermore, after the final whistle, the match officials are obliged to inform the organiser of the competition if the GLT system was used during the match and functioned properly or whether the use of the GLT system was rejected by the referee.

BEFORE KICK-OFF	ACTION	RESPONSIBLE	OPTION
Latest -60 minutes	Final check and decision by the match officials	Referee	Compulsory
	No further adjustments of the GLT system are allowed	Provider of the GLT system	Compulsory
Latest -45 minutes	Information on the final decision given to the stadium operator	Referee	Compulsory
After final whistle	Information to be given to the organiser of the competition on the functionality of the GLT system	Referee	Compulsory

Table 24: Suggested time schedule before kick-off

1.1 Description of referee check

Subject to the confirmation by the GLT provider that the GLT system is ready to use, the match officials must verify that the GLT system is working on both goals and confirm that they will use it for the match.

Depending on the technology employed for the GLT system, the check by the referee has to be carried out with only one or all footballs to be used in the match. This information is to be provided to the match officials at the latest on their arrival in the stadium by the GLT provider or stadium operator.

1.1.1 GLT system without ball modification

When the football has not been modified (no technology inside the ball), it is sufficient for the test to be carried out with one ball. No other ball may be visible on the pitch. If weather conditions are bad, the ball intended for these weather conditions also has to be used in the check by the referee. An international match ball can be identified by the FIFA QUALITY PRO, FIFA QUALITY or "IMS International Matchball Standard" mark.

1.1.2 Test procedures

Three simple tests are applied by the match officials to verify the functionality of the installed GLT system. All referees' watches and the appropriate balls for the GLT system (see C 1.1.1, C 1.1.2) have to be used for the following checks.

1.1.2.1 Test 1

a) Procedure
The ball is to be carried once over the goal line outside the goal.

b) Requirement
No "GOAL" indication on any referees' watches at any time.

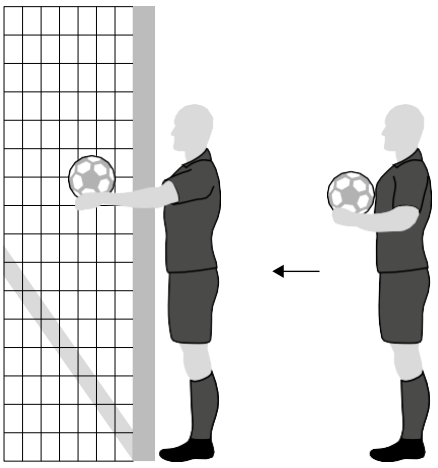


Figure 18: Test 1

1.1.2.2 Test 2

a) Procedure
The ball is to be placed between 3 and 5 metres in front of the goal-line and shot into the goal at moderate speed.

b) Requirement
"GOAL" indication on all referees' watches immediately after the ball has fully crossed the goal-line.

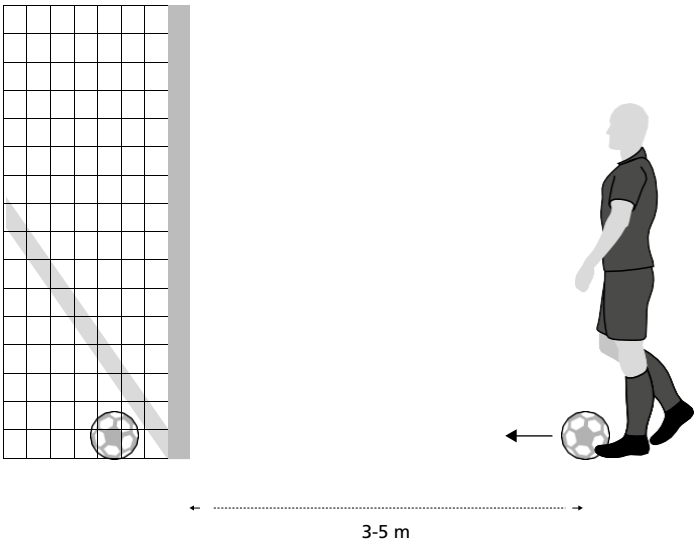


Figure 19: Test 2

1.1.2.3 Test 3

a) Procedure
The ball is held in one hand between 3 and 5 metres in front of the goal-line. The referee then walks into the goal at moderate speed until the ball has physically crossed the line.

b) Requirement
"GOAL" indication on all referees' watches immediately after the ball has fully crossed the goal-line.

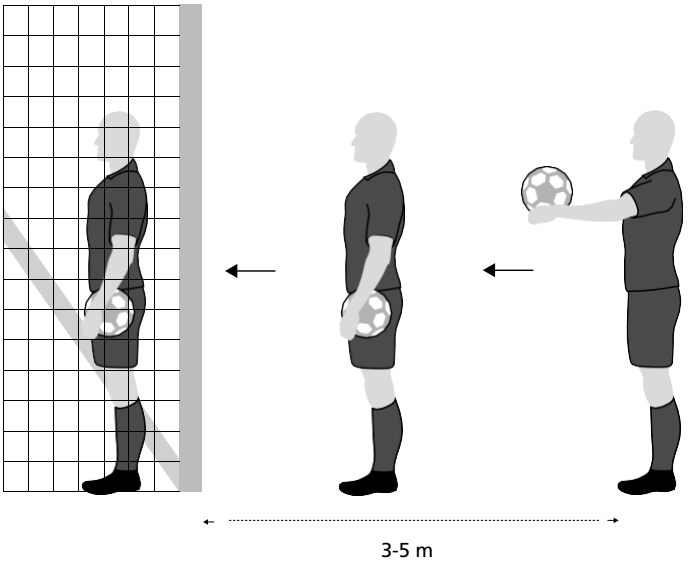
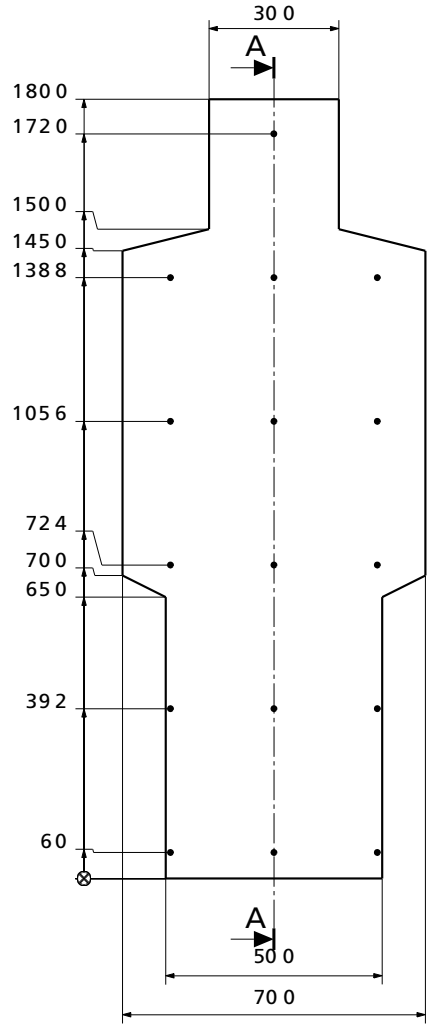


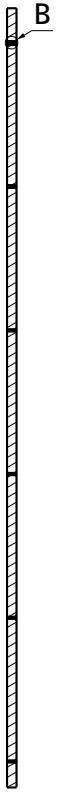
Figure 20: Test 3

2 PROVISION OF INFORMATION TO THE COMPETITION ORGANISER

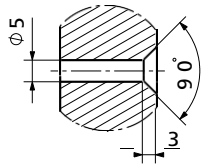
It is recommended that all match officials receive appropriate training in the technology by the competition organiser so that they know how it works before they use it in an official match for the first time. An automatic feedback process should also be set up in order for the organiser of the competition to receive information after the match with regard to how the technology performed.

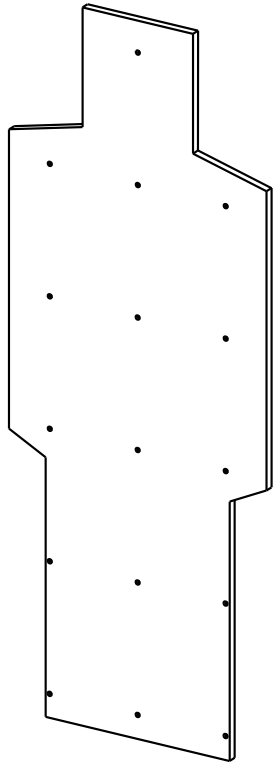


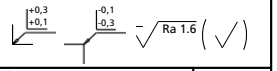
SchnittansichtA-A
Maßstab: 1:10



Detail B
Maßstab: 1:1



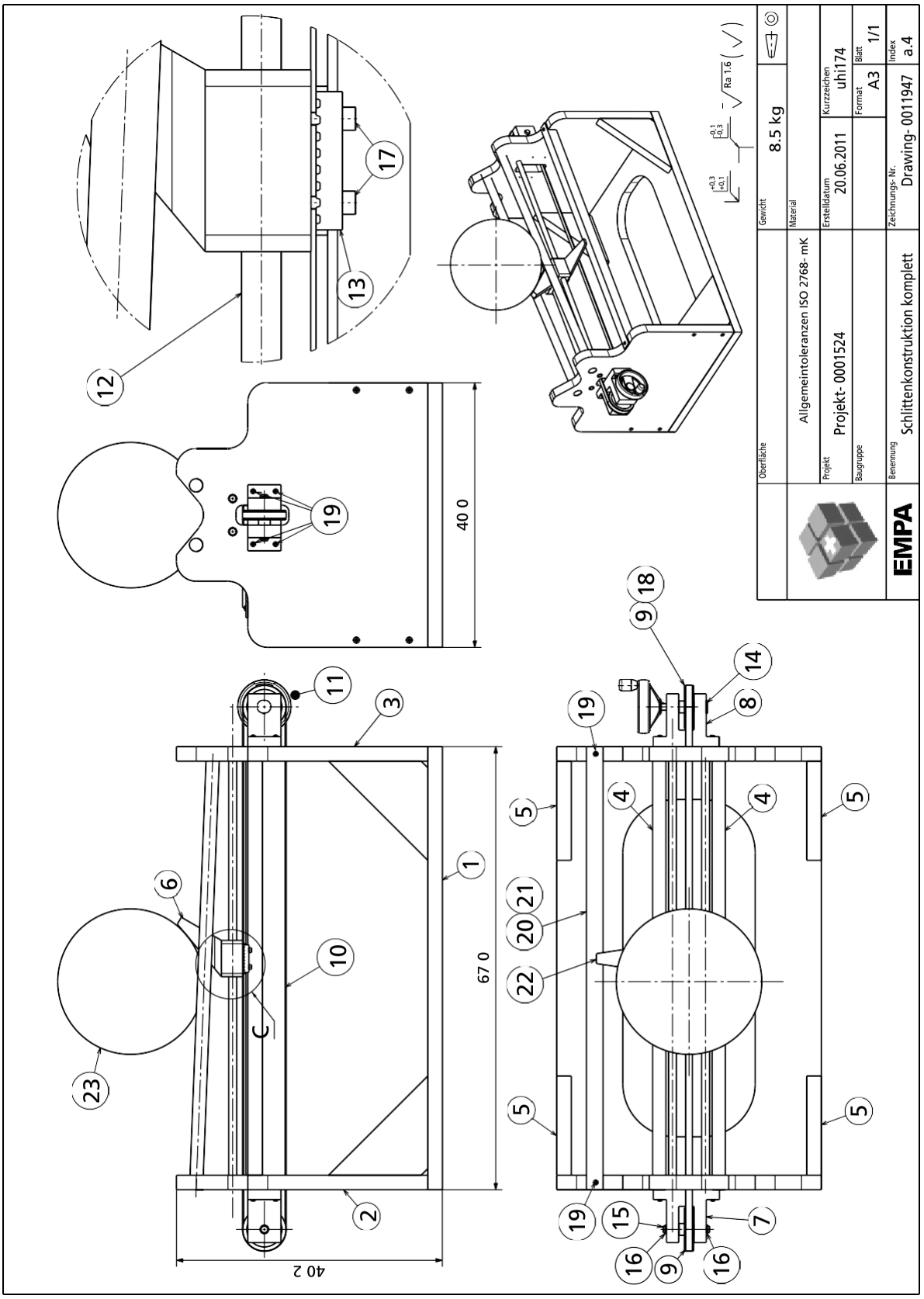




	Oberfläche	Gewicht 15.5 kg	
	Allgemeintoleranzen ISO 2768-mK	Material Sperraform	
	Projekt Projekt-0001524	Erstelldatum 28.06.2011	Kurzzeichen uhi
	Baugruppe Prallwand komplett		Format A3 / Blatt 1/1
EMPA	Benennung Prallwand	Zeichnungs-Nr. Drawing-0011976	Index a.1

For detailed information please contact FIFA (quality@fifa.org).

APPENDIX B: CAD DRAWING OF SLIDER



For detailed information please contact FIFA (quality@fifa.org).

Fédération Internationale de Football Association
FIFA-Strasse 20 P.O. Box 8044 Zurich Switzerland
T: +41 (0)43 222 7777 F: +41 (0)43 222 7907 FIFA.com

